

COMP5339 26s1 Final Practice

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W1 Introduction to Data Engineering and Data Pipelines

Q1. MCQ

A dashboard is updated by a script that downloads a CSV manually, edits column names in a spreadsheet, and uploads the result to a shared folder. Which change most directly turns this into a data-engineering pipeline rather than a manual reporting task?

- (A) Move the spreadsheet to a shared drive but keep the same manual edits.
- (B) Automate ingestion, validation, transformation, and repeatable delivery with logging.
- (C) Add a dashboard refresh button without validating source changes.
- (D) Keep only the final edited spreadsheet and no run logs or validation results.

Q2. Short Answer

Explain the difference between data generation, data acquisition, data transformation, and serving in a typical data pipeline. Use one concrete example.

Q3. Short Answer

A manager says: "If our model is wrong, that is a data-science problem, not a data-engineering problem." Give two ways poor data engineering can create a wrong model conclusion.

Q4. MCQ

Which pair best matches the terms?

- (A) Throughput: delay until data is usable; latency: records processed per second.
- (B) Throughput: records processed per unit time; latency: time from source event to usable output.
- (C) Throughput: number of tables; latency: number of columns.
- (D) Throughput: storage cost; latency: cloud region.

Q5. Short Answer

A pipeline has perfect average throughput but fails during hourly traffic spikes. Explain why average throughput alone is not enough and name two controls.

Q6. Long Question

You inherit a pipeline with no tests, no lineage, and no monitoring. Design a minimal improvement plan that would make failures visible before dashboards become misleading. Include at least four concrete checks or artefacts.

Q7. MCQ

A pipeline reads transaction files nightly. One night the source silently changes the date format and half the rows are rejected. Which quality dimension is most directly threatened?

- (A) Completeness and validity/accuracy.
- (B) Freshness only, because the source file still arrived on time.
- (C) Lineage only, because row rejection cannot affect data values.
- (D) Security only, because any date parsing issue is an access-control problem.

Q8. Short Answer

Why is a pipeline usually evaluated by correctness, efficiency, and ease-of-use rather than speed alone?

Q9. Long Question Semester 2 2023 Final (modified)

A city council wants a dashboard comparing live car-park occupancy with suburb-level public-transport accessibility. Occupancy arrives from a Web API every minute, while accessibility scores are published monthly as CSV files. Propose a high-level data architecture and justify which parts should be streaming, batch, raw-staged, and served to the dashboard.

Q10. Short Answer

Name the four common data-engineering challenge categories volume, variety, velocity, and veracity. For each, give one example risk.



W2 Data Acquisition and Data Cleaning

Q1. MCQ

A source database can emit commit-log changes with operation type and timestamp. Which ingestion strategy is most suitable for a large table that changes frequently?

- (A) Download the whole table every minute.
- (B) Change data capture with idempotent target writes.
- (C) Run a full extract every hour and overwrite the target without operation metadata.
- (D) Append all changed rows without deduplication or delete handling.

Q2. Short Answer

Compare push, pull, and polling ingestion. Give one situation where each is natural.

Q3. Short Answer

A dataset uses blank values, 'NA', -1, and 99999 to represent different kinds of missing values. Why can replacing all of them with zero be dangerous?

Q4. Calculation

A batch pipeline receives 1,200,000 records in 20 minutes. The downstream service can validate 800 records per second on one worker. Ignoring overhead, how many workers are required to keep up with the average arrival rate? Explain the limitation of this calculation.

Q5. Long Question Semester 1 2024 Final (modified)

A hospital vitals feed has 30 percent missing blood-pressure values, and missingness is more common for severe cases because measurements are postponed during emergency treatment. Classify the missingness mechanism and explain an appropriate cleaning strategy.

Q6. MCQ

Which check is most suitable for detecting schema drift before it silently breaks a dashboard?

- (A) Compare incoming field names, types, and required columns against a versioned schema.
- (B) Sort the rows alphabetically.
- (C) Only check whether the final dashboard still has the same number of charts.
- (D) Accept all new fields but never alert on missing required fields.

Q7. Short Answer

Give two examples of data validation rules for a public-transport GPS feed and explain what each catches.

Q8. Long Question

A team cleans customer addresses by lowercasing text and removing punctuation, then joins to postcode boundaries. The join success improves, but some apartment and unit information disappears. Diagnose the bug and propose a safer cleaning workflow.

Q9. MCQ

Which statement about raw staging data is best?

- (A) It is useless once curated tables exist.
- (B) It can support replay, auditing, and changed transformation logic if access is controlled.
- (C) It should always be editable by analysts.
- (D) It removes the need for metadata.

Q10. Short Answer

Explain why data cleaning assumptions should be recorded as metadata or documentation, not just embedded in code.

W3 Databases, SQL, Warehouses, and OLAP

Q1. MCQ

Why can pushing a filter and aggregation into a DBMS be better than exporting the whole table to Python?

- (A) It always changes the answer.
- (B) It can reduce data transfer and use indexes/query optimisation.
- (C) It prevents all data-quality issues.
- (D) It removes the need for SQL permissions.

**Q2. Short Answer**

Distinguish OLTP and OLAP access patterns using an online store example.

Q3. Long Question Semester 1 2024 Final (modified)

A CSV has store, product, sale date, and amount. Design a star schema and explain why it helps analytical queries.

Q4. Short Answer

A finance report changes after an analyst edits a transformation rule. What lineage metadata should be available to audit the change?

Q5. MCQ

Which operation best describes aggregating daily sales to monthly sales in an OLAP cube?

- (A) Roll-up.
- (B) Drill-through to raw logs.
- (C) Packet routing.
- (D) Hash partitioning.

Q6. Calculation

A warehouse table stores 200 million sales rows. A query needs only 4 columns out of 40 and records from one month out of 24 equally sized months. Roughly what fraction of cell values is needed if column pruning and partition pruning both work ideally?

Q7. Long Question Semester 2 2023 Final (modified)

An analytics team wants to build a data lake for raw CSV, JSON API responses, images, and curated warehouse extracts. Explain what makes this a data lake rather than just a folder of files, and evaluate one governance risk if raw data is stored without cataloguing or lifecycle rules.

Q8. Short Answer

What is a slowly changing dimension? Give an example where preserving old dimension values matters.

Q9. MCQ

A dashboard query joins large fact rows to dimensions but one dimension key is missing for many records. Which issue is most likely?

- (A) Referential-integrity or late-arriving dimension problem.
- (B) Successful compression.
- (C) Correct pivoting.
- (D) A late-arriving dimension or inconsistent key mapping.

Q10. Short Answer

Explain why indexes that help OLTP transactions are not automatically enough for OLAP workloads.

W4 Web Scraping and Web APIs**Q1. Long Question Semester 2 2024 Final (modified)**

Design a politeness policy for a small crawler that collects public product pages. Include a simple state flow and how the crawler should handle 4xx errors, 5xx errors, redirects, and timeouts.

Q2. MCQ

Which response should usually cause a crawler to slow down or back off?

- (A) HTTP 429 Too Many Requests.
- (B) HTTP 200 OK.
- (C) A smaller HTML heading.
- (D) A CSV delimiter.

Q3. Short Answer

Compare scraping an HTML page with using an official JSON API for the same data. Give one advantage and one risk of each.

Q4. Short Answer Semester 2 2023 Final (modified)

A scraper is failing because it treats visible article text, navigation menus, and page metadata as one flat string. Sketch the high-level structure of an HTML page and explain why distinguishing the head, body, tags, and nested elements matters for extraction.

**Q5. Short Answer**

Why is a timeout not the same as a negative result from a web API?

Q6. Long Question

A scraper starts returning empty prices after a website redesign. The HTTP status is 200 and pages still contain prices in a browser. Diagnose likely causes and propose monitoring that would catch this earlier.

Q7. Calculation

An API allows an effective steady rate of 600 requests per 10 minutes per token, with no initial burst. You need to fetch 18,000 product records, one request per product, using one token. What is the minimum time ignoring network latency? What design concern remains?

Q8. Short Answer

Give three pieces of metadata a pipeline should store for each API extraction run.

Q9. MCQ

A web API changes one field from integer to string without notice. Which downstream failure is most plausible?

- (A) Type validation or numeric aggregation fails.
- (B) Rate limiting becomes stricter because the field is now quoted.
- (C) A spatial CRS mismatch appears in unrelated distance queries.
- (D) The API authentication token gains broader privileges automatically.

Q10. Short Answer

Explain why authentication secrets for APIs should not be hard-coded in notebooks or committed to version control.

W5 Semistructured Data and NoSQL**Q1. MCQ**

Which property best describes semi-structured data?

- (A) It may have nested, optional, or heterogeneous fields while still carrying structure.
- (B) It cannot be parsed by any program.
- (C) It must be stored only in CSV files.
- (D) It has no labels or tags of any kind.

Q2. Short Answer

Compare XML and JSON for data exchange. Mention one advantage and one disadvantage of each.

Q3. Long Question Semester 2 2024 Final (modified)

A bookstore platform receives XML inventory, JSON user reviews, and scraped HTML book pages. Explain why the data is semi-structured and design a MongoDB collection for books and reviews.

Q4. MCQ

Which MongoDB aggregation approach is most flexible for multi-stage transformations such as match, unwind, group, and project?

- (A) Aggregation pipeline.
- (B) Screenshot aggregation.
- (C) Manual copy-paste.
- (D) HTML heading levels.

Q5. Short Answer

Explain the design trade-off between embedding and referencing in a document database.

Q6. Long Question

A MongoDB collection stores user documents with an unbounded array of click events. Over time some documents exceed practical size and updates slow down. Diagnose the design problem and propose a redesign.

Q7. Short Answer Semester 1 2024 Final (modified)

Briefly compare relational databases and NoSQL databases in terms of data structure and scalability.

Q8. Calculation

A sharded collection stores 90 million events across 9 shards by region. One region accounts for 45 million events. What imbalance risk appears, and what shard-key improvement could reduce it?



Q9. Long Question

Model a recommendation graph with users, books, and authors. Specify node labels, relationship types, and one query to find authors liked by users who follow Alice.

Q10. MCQ

Which statement about NoSQL schema design is best?

- (A) NoSQL removes all need to think about schema.
- (B) The access pattern should strongly influence the document/key/graph structure.
- (C) All relationships should always be embedded.
- (D) Sharding always improves every query.

W6 Temporal Data Engineering

Q1. MCQ

Which pair of time dimensions is needed for bitemporal audit queries?

- (A) Valid time and transaction time.
- (B) CPU time and screen time.
- (C) Latitude time and longitude time.
- (D) HTML time and XML time.

Q2. Long Question

A hospital guideline table has condition, treatment, start date, replacement date, and database load time. Explain why a single last-updated timestamp cannot answer: what was valid on 15 March 2024 according to the database on 30 April 2024?

Q3. Short Answer

Give two reasons time zones can break a data pipeline even when all timestamps look valid.

Q4. Calculation

A sensor emits one reading every 5 seconds per device. There are 240 devices. How many readings arrive per hour? What storage design concern follows?

Q5. MCQ

A stream window includes events by their event timestamp, not arrival timestamp. What problem must the system handle?

- (A) Late and out-of-order events.
- (B) XML declarations.
- (C) Polygon holes.
- (D) Choosing a CRS suitable for measurement rather than only web display.

Q6. Short Answer

Compare point-based and interval-based temporal representation with examples.

Q7. Long Question

An energy provider corrects yesterday's hourly meter readings after a calibration issue. Dashboards already consumed the old values. Design a temporal ingestion strategy that supports correction, replay, and audit.

Q8. MCQ

Which SQL predicate is safest for a half-open validity interval?

- (A) $\text{valid_start} \leq t \text{ AND } t < \text{valid_end}$
- (B) $\text{valid_start} < t \text{ AND } \text{valid_end} < t$
- (C) $t < \text{valid_start} \text{ AND } \text{valid_end} < t$
- (D) $\text{valid_start} = \text{valid_end}$

Q9. Short Answer

What is a watermark in stream processing, and why is it useful for temporal aggregation?

Q10. Long Question

A retailer joins clickstream events to a product catalogue that changes over time. Explain how a temporal join should decide which product category to attach to each click.



W7 Spatial-Temporal Data Engineering

Q1. MCQ

Which geometry type is most natural for a bus GPS observation at one timestamp?

- (A) Point.
- (B) Polygon.
- (C) MultiPolygon boundary.
- (D) HTML body.

Q2. Short Answer

Name suitable geometry types for rainfall gauges, river segments, suburb boundaries, and flood extents.

Q3. Long Question Semester 1 2024 Final (modified)

A spatial pipeline combines Airbnb points in WGS84 with incident polygons in GDA94. Explain the CRS problem and design a query workflow for finding listings within 5 km of incidents.

Q4. Calculation

A GPS feed sends one point every 5 seconds for each of 1,500 buses. Estimate points per day. State one indexing or partitioning strategy.

Q5. Short Answer

Why is computing Euclidean distance directly on latitude and longitude degrees usually wrong?

Q6. MCQ

Which predicate is most appropriate to test whether a point lies inside a suburb polygon?

- (A) ST_Contains or ST_Within.
- (B) ST_Uppercase.
- (C) SQL COUNT only.
- (D) XML standalone.

Q7. Long Question

A prototype stores flood extents as only centroid points. Explain one analysis that becomes impossible or misleading and propose a corrected representation.

Q8. Short Answer

Compare point-based and sequence-based representation for bus movement data.

Q9. MCQ

A spatial index primarily helps by:

- (A) Reducing candidate geometries before expensive exact spatial predicates.
- (B) Converting JSON to XML.
- (C) Proving every polygon is valid.
- (D) Replacing all CRS transformations.

Q10. Long Question

Design a spatial-temporal table for ride-share pickups and drop-offs that supports queries by time window, pickup suburb, and trip distance. Include key fields and two indexes.

W8 Processing Unstructured Data

Q1. MCQ

Which is the best first engineering step before running sentiment analysis over millions of support emails?

- (A) Define ingestion, parsing, metadata, access control, and quality checks.
- (B) Strip message headers before recording source, timestamp, or access metadata.
- (C) Train a model without retaining raw-source references.
- (D) Store only model scores and discard all extraction confidence metadata.

Q2. Short Answer

Give three examples of metadata that make unstructured documents more usable for analytics.



Q3. Long Question

A company extracts text from scanned contracts using OCR. Some pages are rotated, some contain tables, and some contain images. Design a pipeline that prevents low-quality extraction from silently entering a search index.

Q4. MCQ

What is a common reason to convert images or text into feature vectors or embeddings?

- (A) To support similarity search, classification, or retrieval over unstructured content.
- (B) To replace access controls because vectorised data cannot contain sensitive information.
- (C) To remove all data quality checks.
- (D) To make every file smaller than one byte.

Q5. Short Answer

Why can a bag-of-words text pipeline miss important meaning?

Q6. Long Question

A social-media pipeline stores raw posts, extracted entities, and sentiment scores. Later the sentiment model is updated. Explain how to design storage so analysts can compare old and new scores reproducibly.

Q7. Calculation

An embedding index stores 8 million documents with 768-dimensional float32 vectors. Estimate raw vector storage in GB using 4 bytes per float, ignoring index overhead.

Q8. MCQ

Which risk is highest if raw customer emails are copied into a broad analytics workspace?

- (A) Exposure of PII and sensitive content beyond need-to-know access.
- (B) Automatic improvement of data quality.
- (C) Elimination of retention requirements.
- (D) Faster polygon intersection.

Q9. Short Answer

Explain one advantage and one disadvantage of storing only extracted features rather than raw unstructured data.

Q10. Long Question

A news archive search engine returns irrelevant articles because boilerplate navigation text is indexed with article content. Diagnose the pipeline flaw and propose a cleaning strategy.

W9 Stream Data Processing

Q1. MCQ

Which statement best describes a data stream?

- (A) A potentially unbounded, time-varying sequence of records.
- (B) A static table that never changes.
- (C) A PDF cover page.
- (D) A one-row CSV only.

Q2. Short Answer

Why are traditional stored-data DBMS query assumptions often awkward for stream processing?

Q3. Long Question

A parcel platform publishes scan events keyed by parcel id. During a network failure producers retry and a consumer is offline for thirty minutes. Explain how duplicates, ordering, offsets, and retention should be handled.

Q4. MCQ

Kafka preserves order most directly within:

- (A) A partition.
- (B) All topics globally.
- (C) Every consumer group across the cluster.
- (D) Records with the same key after partitioning.

**Q5. Calculation**

A stream processor handles 4,000 events/sec. Input rises to 5,500 events/sec for 10 minutes. Ignoring scaling, how many events of backlog accumulate?

Q6. Short Answer

Explain at-least-once processing and why idempotent writes matter.

Q7. Long Question

A fraud-detection stream joins card transactions to a customer-risk table that updates hourly. Discuss whether the join should use latest processing-time state or event-time state.

Q8. MCQ

Backpressure is used to:

- (A) Prevent overloaded downstream components from being overwhelmed by upstream rate.
- (B) Convert XML attributes to JSON arrays.
- (C) Ensure every stream has no timestamps.
- (D) Delete source metadata.

Q9. Short Answer

Give two examples of stream-monitoring metrics and what each reveals.

Q10. Long Question

A clickstream dashboard counts unique visitors per 5-minute event-time window. Late events arrive up to 20 minutes after the window closes. Propose a windowing and correction strategy.

W10 Scalable Data Engineering**Q1. MCQ**

In Spark, which operation is an action rather than a lazy transformation?

- (A) count()
- (B) select()
- (C) filter()
- (D) withColumn()

Q2. Long Question

A PySpark job reads CSV, selects four columns, filters Australian rows, joins a postcode lookup, then calls count and writes Parquet. Explain lazy evaluation and identify where shuffle or partitioning problems may occur.

Q3. Long Question Semester 2 2023 Final (modified)

A company must count error types across 2 TB of server logs stored across many machines. Explain how a MapReduce-style design decomposes the work, why it can speed up processing, and name one limitation that still needs engineering attention.

Q4. Calculation

A dataset has 1.2 TB across 300 partitions. If one partition has 180 GB and the rest share the remaining data evenly, what scalability problem appears?

Q5. MCQ

Which file format is generally better than raw CSV for repeated analytical scans with column pruning?

- (A) Parquet.
- (B) Plain screenshot.
- (C) HTML with inline styles.
- (D) Uncompressed random text.

Q6. Short Answer

Give two ways to reduce data movement before a distributed join.

Q7. Long Question

A team scales from one worker to twenty workers but runtime barely improves. Give four possible causes and how you would diagnose them.

**Q8. MCQ**

What is the small-files problem?

- (A) Too many tiny files create metadata and scheduling overhead for distributed jobs.
- (B) Small files always compress better.
- (C) Small files eliminate schema drift.
- (D) A file cannot be under 1 MB.

Q9. Short Answer

Explain the difference between horizontal scaling and vertical scaling for a data pipeline.

Q10. Long Question

Design a scalable pipeline for real-time social-media sentiment monitoring. Include ingestion, stream processing, storage, serving, and failure handling.

W11 DataOps and ML Pipelines**Q1. MCQ**

Which DataOps practice most directly helps reproduce yesterday's failed pipeline run?

- (A) Versioned code, configuration, input snapshot references, and run metadata.
- (B) Deleting logs after success.
- (C) Renaming the dashboard.
- (D) Running every step manually.

Q2. Short Answer

What is orchestration in a data pipeline? Give two responsibilities of an orchestrator.

Q3. Long Question

A daily ML feature pipeline silently changes because a source column changes units from dollars to cents. Design tests and monitors to catch this before model scores are served.

Q4. MCQ

Which item is most likely to belong in a data contract?

- (A) Expected fields, types, semantics, freshness, and quality guarantees.
- (B) The current informal column names from one producer's staging table only.
- (C) A promise that consumers never ask questions.
- (D) A producer-side note saying consumers should infer missing fields from context.

Q5. Short Answer

Explain why feature leakage is a data-engineering problem as well as a modelling problem.

Q6. Long Question

A notebook used for a group project becomes a production weekly report. Identify four production risks and propose a DataOps replacement.

Q7. Calculation

A pipeline has three independent validation tasks taking 8, 11, and 15 minutes, followed by a 20-minute aggregation task. What is the minimum runtime if validations run in parallel? What if they run sequentially?

Q8. Short Answer

What is the difference between monitoring data freshness and monitoring data quality?

Q9. MCQ

Which failure mode is a training-serving skew?

- (A) The model is trained on features computed one way but served with features computed differently.
- (B) The warehouse stores raw and curated data.
- (C) A stream has a watermark.
- (D) A polygon has a hole.



Q10. Long Question

Design a lineage record for a feature table used by an ML model. Include enough fields to support audit and rollb

W12 Data Privacy and Security

Q1. MCQ

Which field is most clearly personally identifiable information by itself?

- (A) Full name with date of birth.
- (B) A rounded monthly count by suburb.
- (C) A random row number with no mapping.
- (D) A table name.

Q2. Short Answer

Give three security controls relevant to a data-engineering pipeline handling sensitive customer data.

Q3. Long Question

A bank wants to copy ten years of customer interaction notes into an unrestricted analytics workspace. Identify privacy-by-design problems and propose a safer architecture.

Q4. MCQ

Which statement about de-identification is most accurate?

- (A) It reduces risk but may not eliminate re-identification risk, especially when linked with other data.
- (B) It always makes data public-safe.
- (C) It means deleting the database.
- (D) It only applies to XML files.

Q5. Short Answer

Explain the difference between authentication and authorisation in a data platform.

Q6. Long Question

A health analytics team wants suburb-level disease dashboards. Small suburbs sometimes have counts of one or two. Explain the privacy risk and design a release control.

Q7. Calculation

A dataset has 2,000,000 rows. A retention policy requires deleting records older than 7 years. If 18 percent are older than 7 years, but 40,000 of those expired rows are under legal hold and must be retained, how many rows are deleted now?

Q8. Short Answer

Why are audit logs themselves sensitive data?

Q9. MCQ

Which approach best follows least privilege?

- (A) Give analysts only the views and fields needed for their approved task.
- (B) Give everyone admin access because it is faster.
- (C) Grant warehouse-wide read access to all analysts for convenience.
- (D) Disable all logs.

Q10. Long Question

A data lake stores raw API responses containing IP addresses and precise mobile locations. Design a governance policy that balances analytics value with privacy and security.



Answer Key

W1 Introduction to Data Engineering and Data Pipelines

- Q1. B. The key shift is repeatable automated data movement with validation, transformation, logging, and delivery.
- Q2. Data generation is where events or observations originate. Acquisition brings them into the platform. Transformation cleans, reshapes, joins, or enriches them. Serving exposes curated data to analytics, dashboards, ML, or downstream systems. Example: bus GPS devices generate observations, an API or message broker ingests them, a job maps coordinates to routes and removes invalid points, then a dashboard queries route-delay aggregates.
- Q3. Poor engineering can create missing, stale, duplicated, biased, or schema-broken data that the model treats as real. It can also lose lineage and quality metadata, so analysts cannot see when a training set mixes incompatible periods, sources, or definitions.
- Q4. B. Throughput is rate; latency is delay from event/source to usable output.
- Q5. Average throughput can hide burst overload. A pipeline must handle peak arrival rates and backpressure. Controls include buffering/queues, autoscaling workers, rate limits, partitioning, load shedding for non-critical data, and lag monitoring.
- Q6. A good plan includes data freshness checks, row-count and volume anomaly checks, schema and type validation, null, range, and business-rule checks, source-to-target reconciliation, lineage metadata, versioned transformation code, alerting on failed jobs or excessive lag, and a run log with source versions. The answer should prioritise visibility and reproducibility rather than only faster code.
- Q7. A. The changed format causes valid records to be rejected, so completeness is reduced and validation/accuracy assumptions are broken.
- Q8. Speed without correctness gives fast wrong answers. Efficiency matters for cost and scalability, but ease-of-use affects maintainability, reproducibility, onboarding, and whether analysts can safely consume outputs. A good pipeline balances these properties under operational constraints.
- Q9. A strong answer separates live events from slower reference data: streaming or micro-batch ingestion for car-park occupancy, scheduled batch ingestion for monthly CSV accessibility scores, raw immutable staging for replay/audit, curated relational or warehouse tables keyed by location/time, validation and lineage for both sources, dashboard-serving aggregates, monitoring, and access controls. It should also mention different freshness requirements for live occupancy and monthly reference data.
- Q10. Volume: storage/compute bottlenecks. Variety: incompatible formats or schemas. Velocity: late or out-of-order events and latency pressure. Veracity: missing, incorrect, duplicated, biased, or inconsistent data.

W2 Data Acquisition and Data Cleaning

- Q1. B. CDC avoids repeated full extracts and can preserve operation order and metadata when designed correctly.
- Q2. Push: source sends data when events occur, natural for webhooks or message brokers. Pull: target requests data on demand, natural for APIs or files. Polling: target checks periodically for changes, natural when a source lacks event notifications but exposes an updated endpoint or directory.
- Q3. Zero may be a real value and changes distributions, aggregates, and model behaviour. Different placeholders may have different meanings, so they should be mapped to proper missing values or domain categories after validation, with assumptions documented.
- Q4. Arrival rate is $1,200,000 / 1,200 = 1,000$ records/s. Two workers are required because one handles 800 records/s. The calculation ignores bursts, retries, skew, I/O overhead, and downstream latency, so a production design would include headroom and monitoring.
- Q5. This is likely MNAR or at least not MCAR because missingness depends on severity and possibly the unobserved value itself. A good strategy flags missingness, investigates workflow causes, uses domain-aware imputation only with caution, reports bias risk, avoids silently dropping severe cases, and may model missingness explicitly.
- Q6. A. Schema validation catches renamed, missing, added, and type-changed fields.
- Q7. Examples: latitude/longitude must be within plausible NSW bounds; timestamp must not be in the future or far before ingestion; bus id must match known fleet ids; speed derived from consecutive points must not exceed a plausible maximum. Each catches invalid coordinates, time errors, unknown entities, or impossible movement.
- Q8. The cleaning destroys meaningful address tokens such as unit numbers or separators. A safer workflow preserves raw values, normalises into separate parsed fields, uses address-standardisation libraries or reference data, tracks match confidence, stores rejected/ambiguous records for review, and documents transformations.
- Q9. B. Raw staging supports replay and audit, but must be governed and documented.
- Q10. Future users need to know what was removed, imputed, transformed, or treated as invalid. Documentation supports auditability, reproducibility, debugging, and fair interpretation of downstream results.



W3 Databases, SQL, Warehouses, and OLAP

- Q1. B. DBMS execution can reduce transfer and exploit indexes, statistics, and query planning.
- Q2. OLTP handles frequent small reads/writes such as placing orders and updating inventory. OLAP scans at large historical data for analysis such as monthly revenue by region and product category. Their schema and priorities differ.
- Q3. FactSales should store numeric measures such as amount and foreign keys to StoreDim, ProductDim, DateDim, and possibly PromotionDim. Dimensions store descriptive attributes and hierarchies. The star schema simplifies joins, supports roll-up/slice/dice by dimensions, and separates measures from descriptors.
- Q4. Source versions and extraction times, transformation code version, rule parameters, job run id, input/output table versions, user or service account, validation results, and dashboard query/version should be available.
- Q5. A. Roll-up moves to a coarser level in a hierarchy.
- Q6. It needs 4/40 of columns and 1/24 of month partitions, so $1/10 * 1/24 = 1/240$ of cell values, ignoring metadata and uneven sizes.
- Q7. A data lake is a managed repository for diverse raw and processed data, including structured, semi-structured, and unstructured data. It should include metadata/catalogue entries, access control, lineage, schema-on-read or curated zones, and lifecycle policies. Without governance it becomes a data swamp: users cannot find trusted data, privacy controls weaken, and reports may use stale or undocumented files.
- Q8. An SCD tracks changes to dimension attributes over time. Example: a customer's region or a product category changes; preserving historical values allows old sales to be reported using the classification that was valid then.
- Q9. A. Missing keys indicate referential-integrity, data-quality, or late-arriving dimension issues.
- Q10. OLAP queries often scan, group, and join large portions of history across many dimensions. They may benefit from columnar storage, partitioning, materialised aggregates, and distributed execution rather than row-level point-lookup indexes alone.

W4 Web Scraping and Web APIs

- Q1. A good answer includes robots/terms awareness where applicable, per-host rate limits, request queues, backoff, retry caps, user-agent identification, redirect limits, permanent failure for many 4xx cases, retry/backoff for transient 5xx/timeouts, logging, and a circuit breaker for repeated failures.
- Q2. A. HTTP 429 indicates rate limiting; the crawler should back off according to headers or policy.
- Q3. An API is usually more structured, stable, documented, and permissioned, but can have quotas and authentication constraints. HTML scraping can access data without an API but is brittle, may violate terms, and mixes content with presentation.
- Q4. A good answer identifies HTML as tag-based text markup with nested elements, commonly structured with html, head, title or metadata, and body sections. Extraction should distinguish body content from metadata/navigation, preserve useful hierarchy such as headings, links, lists, and tables, and avoid treating all visible and non-visible text as equivalent.
- Q5. A timeout means the client did not receive a response within the limit; the server may still process the request or the network may be slow. A negative result is a valid response saying no data matched. Pipelines should log, retry, and distinguish these states.
- Q6. Likely causes include changed CSS selectors, JavaScript-rendered content, bot detection, locale/currency changes, or hidden markup differences. Monitoring should include extraction yield, null-rate alerts, sample page snapshots, schema-selector tests, canary URLs, and manual review queues.
- Q7. Assuming a steady effective rate with no initial burst, 600 per 10 minutes is 60 per minute. $18,000 / 60 = 300$ minutes, or 5 hours. Remaining concerns include quotas, retries, pagination, data freshness, backoff, and whether bulk endpoints or incremental sync are available.
- Q8. Examples: endpoint and parameters, request time window, response status, pagination cursor, token/client id, schema version, row count, checksum, error count, and source timestamp.
- Q9. A. Type changes can break validation, parsing, joins, and numeric calculations.
- Q10. Secrets in code can leak to collaborators, logs, or repositories. They should be stored in managed secret stores or environment variables with rotation, least privilege, and audit logging.

W5 Semistructured Data and NoSQL

- Q1. A. Semi-structured data is irregular but still has tags, keys, nesting, or other structure.
- Q2. XML supports attributes, namespaces, validation schemas, and mixed content but is verbose. JSON is compact and maps naturally to web APIs and document databases but has fewer built-in semantic constraints and can suffer from loose typing or schema drift.
- Q3. A strong design recognises nested and optional fields. A book document may include ISBN, title, authors, categories,



- prices, and source metadata, with an embedded bounded sample of reviews or separate review documents if review volume is large. The answer should justify embedding versus referencing and include source and schema.
- Q4. A. The aggregation pipeline composes stages such as match, unwind, group, and project.
- Q5. Embedding improves locality and atomic access for data read together, but can duplicate data, create large documents, and make many-to-many updates harder. Referencing avoids duplication and supports large relationships but requires additional queries or joins/lookup operations.
- Q6. The unbounded array causes document growth, write contention, and size/performance problems. Redesign by storing events in a separate collection partitioned by user/time, using buckets, summarising old events, indexing query fields, and retaining user profile data separately.
- Q7. Relational databases use fixed schemas, tables, SQL, constraints, and strong consistency patterns. NoSQL systems vary: document, key-value, wide-column, graph. They often favour flexible schemas and horizontal scalability, but may shift consistency, joins, or validation responsibilities to the application.
- Q8. One shard or region group can become a hotspot with about half the data/workload. A better shard key may combine region with hashed user id or time bucket, depending on query patterns, to distribute load while preserving useful locality.
- Q9. Nodes may include User, Book, Author. Relationships: FOLLOWS, REVIEWED or LIKES, WROTE. Example: MATCH (:User name:'Alice')<-[:FOLLOWS]-(u:User)-[:LIKES]->(:Book)<-[:WROTE]-(a:Author) RETURN a, count(*) ORDER BY count(*) DESC. Variants are acceptable if direction and semantics are clear.
- Q10. B. NoSQL design is usually access-pattern driven and still needs schema discipline.

W6 Temporal Data Engineering

- Q1. A. Valid time records when a fact is true in the real world; transaction time records when the database knew it.
- Q2. A single timestamp cannot distinguish real-world validity from when the record was stored or corrected. The design needs valid-time intervals and transaction-time intervals. The query filters `valid_start <= 15 Mar 2024 < valid_end` and `tx_start <= 30 Apr 2024 < tx_end`.
- Q3. Different sources may use local time, UTC, daylight-saving rules, or ambiguous offsets. Joins, windows, daily aggregates, and ordering can become wrong unless timestamps are normalised and original timezone metadata is preserved.
- Q4. Each device emits 720 readings/hour. 240 devices emit 172,800 readings/hour. The design needs partitioning by time/device, compression/retention policies, and efficient time-range queries.
- Q5. A. Event-time windows must handle late and out-of-order arrivals.
- Q6. Point-based representation stores individual instants such as a GPS reading at a timestamp. Interval-based representation stores facts valid over periods such as a price from `start_date` to `end_date`. Intervals support duration and overlap queries.
- Q7. A strong design stores immutable raw events, versioned corrected records, source event time, ingestion/transaction time, correction reason, and job run ids. Curated tables should support upserts or bitemporal versions, dashboards should read current-as-of views, and audit queries should reconstruct previous knowledge states.
- Q8. A. Half-open intervals avoid double-counting boundary instants.
- Q9. A watermark is a system's estimate that no earlier event-time records, or very few, will arrive. It lets the engine close windows and emit results while still allowing a defined lateness policy.
- Q10. The join should use the click event time to select the catalogue version whose valid interval includes that time, not simply the latest category. If auditing is required, transaction time may also constrain what catalogue version was known when the report ran.

W7 Spatial-Temporal Data Engineering

- Q1. A. A single observed location is a point.
- Q2. Rainfall gauges: Point. River segments: LineString or MultiLineString. Suburb boundaries: Polygon or MultiPolygon. Flood extents: Polygon or MultiPolygon.
- Q3. The data uses different coordinate reference systems, so direct comparison can misplace features or compute wrong distances. Transform both layers to a common appropriate projected CRS for distance, create spatial indexes, then use predicates such as ST_DWithin or buffered intersection. Store original CRS metadata and validate transformations.
- Q4. Each bus sends 17,280 points/day. Total is 25,920,000 points/day. Use time partitioning plus spatial index such as GiST, possibly partition by date and route/vehicle id.
- Q5. Degrees are angular units, not equal linear distances everywhere. Longitude degree length changes with latitude, and Earth's curvature matters. Transform to an appropriate projected CRS or use geodesic functions.
- Q6. A. Containment predicates test point-in-polygon relationships.
- Q7. Centroids cannot represent area, boundaries, overlap percentages, or partial coverage. A suburb may have a centroid outside a flood zone while much of its area is flooded. Use Polygon/MultiPolygon geometries with CRS metadata and area-based predicates such as intersection area divided by suburb area.



- Q8. Point-based stores each GPS observation independently and is simple for recent-location queries. Sequence-based stores trajectories or ordered line segments and supports route-shape and movement analysis, but can be harder to query incrementally.
- Q9. A. Spatial indexes filter candidate bounding boxes before exact checks.
- Q10. A good schema includes trip_id, vehicle/user pseudonym, pickup_time, dropoff_time, pickup_geom, dropoff_geom, optional trajectory or distance, CRS, and source metadata. Indexes: time partition/index on pickup_time; GiST/SP-GiST spatial indexes on pickup and dropoff geometries; possibly composite partitioning by date.

W8 Processing Unstructured Data

- Q1. A. Unstructured pipelines still need governed ingestion, metadata, quality checks, and access control.
- Q2. Examples: source system, capture time, author or owner, language, document type, encoding, access classification, extraction confidence, OCR version, checksum, and lineage id.
- Q3. A strong design includes preprocessing, page orientation detection, OCR confidence thresholds, table extraction checks, manual review queues, raw image retention, extracted-text versioning, metadata, sampling audits, and alerts when low-confidence rate changes.
- Q4. A. Embeddings/features make unstructured content usable for similarity and ML tasks.
- Q5. It may ignore word order, negation, context, sarcasm, domain-specific meanings, and document structure. It can also overweight frequent but uninformative terms if not designed carefully.
- Q6. Store raw immutable posts, model version, feature/extraction version, scoring time, parameters, and output tables versioned by model. Allow backfills into new score columns/tables rather than overwriting without history. Keep lineage from raw post to entity extraction and score.
- Q7. $8,000,000 * 768 * 4 = 24,576,000,000$ bytes, about 24.6 GB decimal or about 22.9 GiB. Real indexes need additional overhead.
- Q8. A. Raw unstructured text often contains sensitive or identifying information.
- Q9. Features reduce privacy/storage exposure and may speed analytics, but lose context, make reprocessing difficult, and can preserve model/extraction errors. Keeping raw data may support audit and improved extraction but requires stricter governance.
- Q10. The extraction did not separate main content from page template. Use boilerplate removal, DOM-aware extraction, content-density heuristics, deduplication of repeated blocks, fielded indexing for title/date/body, validation samples, and source-specific parser tests.

W9 Stream Data Processing

- Q1. A. A stream is potentially unbounded and evolves over time.
- Q2. DBMS queries usually run over finite stored data. Streams are unbounded, arrive continuously, may be late/out of order, and require low-latency incremental computation with windows, state, and backpressure.
- Q3. Use parcel id as key for per-parcel ordering within partitions, idempotent producers/consumers, event ids or parcel id plus scan timestamp for deduplication, commit offsets only after durable processing, sufficient retention for offline consumers, retry/dead-letter topics, monitoring lag, and replay procedures.
- Q4. A. Kafka ordering is per partition, not globally across all partitions.
- Q5. The deficit is 1,500 events/sec for 600 seconds, so backlog grows by 900,000 events.
- Q6. At-least-once means a record should not be lost but may be processed more than once after retries or failures. Idempotent writes make repeated processing produce the same final state, preventing duplicate effects.
- Q7. Latest state is simpler and may fit real-time decisions, but historical replay can produce different results and may be unfair for old events. Event-time state supports reproducible historical decisions but needs versioned state and temporal joins. The answer should match the business/audit requirement.
- Q8. A. Backpressure manages flow when downstream cannot keep up.
- Q9. Consumer lag reveals whether processing falls behind. Throughput and latency show processing capacity and freshness. Error/dead-letter rates show bad records or downstream failures. Watermark delay shows lateness.
- Q10. Use event-time windows with a watermark and allowed lateness. Emit preliminary results, update results when late events arrive within the allowed window, and send very late records to correction/reconciliation. Store window version metadata so dashboards know whether results are final.

W10 Scalable Data Engineering

- Q1. A. count() triggers execution; the others define transformations.



- Q2. Transformations build a logical plan until actions such as count or write execute it. Filter and projection are pushed down/pruned where possible. The join may cause a shuffle unless the lookup is broadcast or data is partitioned. Skewed postcodes can overload tasks; optimisations include broadcast joins, repartitioning, salting skewed data, pruning, and using Parquet.
- Q3. A MapReduce-style design maps independent log partitions to partial key-value counts such as error_type, shuffles/groups by key and reduces partial counts into global totals. It speeds processing by parallelising work near distributed data and combining partial results. Limitations include shuffle cost, skewed keys, fault tolerance, straggler tasks, and the fact that not every pipeline step is naturally MapReduce-friendly.
- Q4. The 180 GB partition is a skewed straggler. Even if most partitions are about 3.4 GB, the large partition can dominate runtime and cause memory/disk failures.
- Q5. A. Parquet is columnar and supports efficient analytical scans.
- Q6. Filter early, select only needed columns, broadcast a small table, partition/bucket by join key, pre-aggregate, or use data locality-aware storage layouts.
- Q7. Possible causes: serial bottleneck, shuffle-heavy plan, skewed key, small-file overhead, insufficient partitions, slow source/sink, driver bottleneck, memory spills, or network limits. Diagnose with execution plans, stage/task metrics, spill/shuffle counters, partition size distribution, source/sink throughput, and cluster utilization.
- Q8. A. Many tiny files make listing, scheduling, and metadata handling inefficient.
- Q9. Vertical scaling increases resources on one machine. Horizontal scaling adds more machines/workers and requires partitionable work, coordination, and distributed storage/processing.
- Q10. A strong answer uses API/stream ingestion into a broker, stream processors for parsing/language filtering/model scoring, scalable storage for raw and curated data, feature/model version metadata, dashboard-serving aggregates, retries/dead-letter topics, lag/error monitoring, privacy filtering, and autoscaling/backpressure.

W11 DataOps and ML Pipelines

- Q1. A. Reproducibility needs versioned artefacts and run metadata.
- Q2. Orchestration schedules and coordinates tasks with dependencies. Responsibilities include triggering jobs, ordering DAG steps, retries, parameter passing, logging, alerting, and tracking task state.
- Q3. Use schema/type checks, unit/range expectations, distribution drift checks, source metadata validation, feature summary comparison against baselines, canary scoring, model input contracts, alerts, and blocking rules for severe anomalies.
- Q4. A. Data contracts define expected structure, meaning, freshness, and quality/service expectations.
- Q5. Pipelines may accidentally join future information, post-outcome fields, or labels into training features. Engineering controls around temporal joins, data contracts, and training/serving parity are needed to prevent leakage.
- Q6. Risks include hidden state, manual execution, unpinned dependencies, no tests, no logging, no permissions model, unreproducible inputs, and fragile paths. Replace with version-controlled jobs, environment management, orchestrated DAGs, tests, monitoring, secrets management, and documented outputs.
- Q7. Parallel runtime is $\max(8, 11, 15) + 20 = 35$ minutes. Sequential runtime is $8 + 11 + 15 + 20 = 54$ minutes.
- Q8. Freshness checks whether data arrived or was updated within an expected time. Quality checks whether values, schema, distributions, completeness, and business rules are acceptable.
- Q9. A. Training-serving skew occurs when training and production feature generation differ.
- Q10. Include feature table version, source table versions, extraction windows, transformation code commit, parameters, job run id, validation results, schema version, owner, model version using it, creation time, and retention/access classification.

W12 Data Privacy and Security

- Q1. A. Full name with date of birth can identify a person.
- Q2. Examples: least-privilege access control, encryption in transit and at rest, secrets management, network isolation, audit logging, data masking, MFA, key rotation, vulnerability patching, and incident monitoring.
- Q3. Problems include excessive collection/use, broad access, retention risk, sensitive unstructured notes, and unclear purpose limitation. Safer design uses data minimisation, retention rules, restricted raw zone, role-based access, pseudonymisation/tokenisation, redaction of sensitive notes, aggregated outputs, audit logs, encryption, and approval workflows.
- Q4. A. De-identification must be assessed against linkage and context risk.
- Q5. Authentication verifies who or what the user/service is. Authorisation determines what that identity is allowed to access or do.
- Q6. Small cells can reveal or imply individual health status, especially when combined with external knowledge. Controls include aggregation to larger regions/time periods, small-cell suppression, noise/privacy-preserving methods, access tiers, and review before release.
- Q7. Expired rows = 360,000. Legal hold overrides deletion for 40,000 of those rows, so deleted rows = $360,000 - 40,000 =$



320,000. The remaining retained rows are 1,680,000.

- Q8. Audit logs can reveal user identities, access patterns, dataset names, queries, IP addresses, and incident details. Integrity protection, retention rules, and restricted access.
- Q9. A. Least privilege limits access to what is required for the task.
- Q10. A strong policy classifies raw data as sensitive, restricts raw-zone access, encrypts and logs access, minimises retention, aggregates or coarsens location for most analytics, pseudonymises identifiers, separates key mapping, documents lawful purpose, applies deletion/erasure workflows, and reviews outputs for re-identification risk.